# 13 Digital Sum: 9's & 3's Reminder

| 9's | 3's |
|---|---|
| 9) Any Num ( | 3) Any no ( |
| R = digital sum | R = Baby digital sum |
| for 9 the R is | for 3 the R is |
| R = 1, 2, 3 ---- 8 | R = 0, 1, 2, ~~3~~ |

R = 0 — 81 ÷ 9
— 9 ÷ 9

R = 0

Ex:- $\cancel{3}\ \underbrace{2\ 1}_{3,\ R=0}$, R = 0

Ex:

321 = 3 + 2 + 1 = 6

2358 = 2 + 3 + 5 + 8 = 18 = 1 + 8 = $\boxed{9}$ = 0

→ 9's & sum of digits that give 9 should be ignore.

Ex:- $\underbrace{\cancel{2}\ \cancel{7}}_{\cancel{9},\ R=0}\ 2\ 3\ \underbrace{\cancel{9}}_{R=0}$ = 2 + 3 = 5

*) for division the last not be the

* $\overline{3}$, $\overline{6}$, $\overline{9}$ (to see not come this number in division

Ex:- $\frac{27}{9}$ for this situation

$\frac{27 \div 9}{9 \div 9} = \frac{3}{1}$

Question :-

1) $1.5 + 32.5 + 23.9 = ?$

Sol:

2) $321 + 467 + 2311 = ?$ find sum of digit

Sol:

digital sum is:

1. sum of digits
2. sum of digit to single digit.

$\underset{6}{\underline{321}} + \underset{17}{\underline{467}} + \underset{7}{\underline{2311}}$

$6 + 8 + 7 = 21 = 2+1 = \underline{3}$

option

a) ~~4000~~ ×

b) ~~5000~~ ×

c) ~~6000~~ ×

d) ~~3000~~ ✓ because digital sum is $\underline{3}$

1) $1.5 + 32.5 + 23.9 = ?$

$\underline{6} + \underline{10} + \underline{14}$

$6 + 1 + 5 = 12 = 1+2 = 3$

a) $67.3 = 16 = 1+6 = 7$ ×

b) $57.9 = 21 = 2+1 = 3$ ✓

c) $37.6 = 16 = 1+6 = 7$ ×

d) $47.8 = 19 = 1+9 = 10 = 1+0 = 1$ ×

(3) digital sum of 93 - 21 ?

Sol^n

= $\cancel{9}3 - 21$

= 3 - (2+1)

= 3 - 3 = 0

or,

= 93 - 21

= 72

= 7 + 2 = 9

R = 0.

(A) Top questions

(4) 637.28 - 781.47 + 257.30 = ?

= 8 - 0 - 8

= 16 = 1 + 6 = 7

a) 113.2 = 7 Ans ✓

(5) 53% of 120 + 25% of 862 = ?

% of 500

Sol^n

53% of 120 + 25% of 862 = ? % of 500

= 8 × 3 + 7 × 7 = x % × 5

ignore

= 24 + 4$\cancel{9}$ = 5x

= 6 + 4 = 5x

10 = 5x

x = $\frac{10}{5}$ 2

x = 2 ✓

or

1 + 0 = 5x

1 = 5x

x = $\frac{1}{5}$ = $\frac{1 \times 2}{5 \times 2}$ = $\frac{2}{10}$ =

→ Alwa… decima… 10…

55.82

20 = 2 + 0 =

Ans = $\boxed{55.82}$

(6) If $x = 2$, find $x^3 + 27x^2 + 243x + 631$

Sol^n

$x^3 + \underset{9}{\underline{27}}\, x^2 + \underset{9}{\underline{243}}\, x + 631$

R=0 R=0

$= x^3 + \cancel{0(x^2)} + \cancel{0(x)} + 1$

$= x^3 + 1$

$= (2)^3 + 1$

$= 8 + 1 = 9 = 0 \; (R = 0)$

Ans $\underline{1233}$ Add (9)

(7) If $x = 2$, $y = 9$ &

$A = \dfrac{x^8 - 1}{x^4 + 1}$ $B = \dfrac{y^4 - 1}{y^2 + 1}$ then.

find $A^2 + 2AB + AB^2$.

Sol^n w.k.T

$a^2 - b^2 = (a+b)(a-b)$

$\therefore A = \dfrac{x^8 - 1}{x^4 + 1}$

$A = \dfrac{(x^4 - 1)\cancel{(x^4 + 1)}}{\cancel{(x^4 + 1)}}$

$A = x^4 - 1$

$A = (2)^4 - 1 = 16 - 1 = 15 = 1 + 5 = 6.$

$A = 6.$

$B = \frac{y^{2}-1}{y^2+1}$

$B = \frac{(y^2-1)\cancel{(y^2+1)}}{\cancel{(y^2+1)}}$

$B = y^2-1 = (9)^2-1 = 81-1 = 80 = 8+0 = 8$

$B = 8$

$\therefore$ Given :- $A^2 + 2AB + AB^2$

$(6)^2 + 2(6)(8) + 6(8)^2$

$\cancel{3}6 + 2\times 6\times 8 + 6\times 64$

$= 0 + \cancel{9}6 + 6\times(\cancel{63}+1)$

$= 0 + 6 + 6\times(1)$

$= 12 = 1+2 = \underline{3}$

Ans :-

a) 8.8125 = 7 ✗

b) 82425 = 4 ✗

c) 986.25 = 21 = 2+1 = $\underline{3}$ ✓

d) 96475 = 22 = 2+2 = 4 ✗

Ans :- $\underline{986.25.}$

(8) $(1004)^2 - (998)^2 = ?$

Sol$^n$

$= (1004)^2 - (998)^2$

$= (5)^2 - (8)^2$

$= 25 - 64$

$= 7 - 10$

$= 7 - 1$

$= \underline{6}$

Ans : $\underline{12012}$

12012

$\underline{6}$

(9) If $A = \frac{1}{0.4} + \frac{1}{0.04} + \frac{1}{0.004} + \cdots$ 8 terms

Sol$^n$

$A = \frac{1}{0.4} = \frac{1}{\frac{4}{10}} = \frac{\cancel{10}^5}{\cancel{4}_2} = \frac{5}{2} = 2.5$

$= 2.5 = 2+5 = 7.$

$\therefore A = 7+7+7+ \cdots$ 8 terms

$A = 7(8) = 56 = 11 = \underline{1+1}$

$A = 2$

Ans $\underline{27777777.5}$

Sol$^n$ $\cancel{2}\cancel{7}777777.5$

$\cancel{8}$

$42+5$

$= 6+5 = 11 = \underline{1+1} = \underline{2}$

(9) find Value of

$= \underbrace{(1,000,001)^2}_{+} - (\cancel{999,999})^2$

Sol$^n$

$= (2)^2 - (0)^2$

$= 4.$

Ans $\underbrace{4,000,000.}_{+} \rightarrow \underline{4}$

(10) which minimum should be add to 13851 so that it will be divided by 87 exactly.

Sol$^n$ Given

$$\frac{13851 + (x)}{87} = \frac{\quad}{8+7} = \frac{\quad}{15/3} = 4\,4$$

(use 3's)

Ans (C) 69, which is divisible by 3's.

(11) 375375 ÷ 455 + 13.3% of 8600 − 15.7% of 9240 = 40% of x find the Value of 'x'

Sol$^n$

$$\frac{\overbrace{375375}^{+27=9}}{\underset{9}{455}} = \frac{3}{5} = \frac{3\times 2}{5\times 2} = \frac{6}{10} = \frac{6}{1}$$

digital sum = $\underline{6}$

for dinominator always comes to 10 is

the dinominator 3, 6, & 9, this are not possible. the possibles are

| | | | | | |
|---|---|---|---|---|---|
| 1 × 1 | 2 × 5 | 4 × 7 | 5 × 2 | 7 × 4 | 8 × 8 |
| | | 28 + 10 | 10 | 28 + 10 | 64 + 10 |

mul →

sol[2]

$13.3\% \text{ of } 8600 - 15.7\% \text{ of } 9240$

$7\% \times 14 \qquad 13\% \times 6$

$7 \times 5 \qquad 4 \times 6$

$\frac{35}{8} \quad - \quad 24$

$8 - 6$

for earlier. $\frac{375375}{455}$

$\frac{375375}{455} + 13.3 \times 8600 - 15.7 \times 9240$

$\cancel{6} + 8 - \cancel{6}$

11

$8 = 40\% x$

$8 = 4x$

$x = \frac{8^2}{4}$

$x = 2$ digital sum

Ans = 1295.3 ← comes.

(14) Find the value of :-

$$+\frac{6410.25}{3465}+(1.12\times1.18\times1.25)+(1.392\%\text{ of }1605)+(1.4)^3-(1.48)^2$$

$\hookrightarrow$ (1).

$\frac{6410.25}{3465} = \frac{9}{18} = \frac{9}{9}$ w.k.t 9 dont come divide side.

$\therefore \frac{6410.25 \div 9}{3465 \div 9} = \frac{712.25}{385}$

Apply digital sum.

digital sum

$\frac{712.25}{385} = \frac{8}{16} = \frac{8}{7} = \frac{8\times4}{7\times4} = \frac{32}{28} = \frac{5}{10}$.

$= \underline{5}$

for

$= 1.12 \times 1.18 \times 1.25$

$= 4 \times 1 \times 8$

$= 32 = \underline{5}$ (digital sum).

for

$= 1.392\ \%\ \text{of}\ 1605$

$= 1.392\ \%\ \times\ 1605$

$= 6 \times 12$

$= 6 \times 3$

$= 18 = \cancel{9}$ (digital sum)

$= \underline{0}$

for

$(1.4)^3 - (1.48)^2$

$(5)^3 - (1)^2$

$125 - 1$

~~$= 124 = 8$~~

$= 8 - 1 = 7$ (ds).

25
25
125

put in eq^2 (1)

$= 5 + 5 + 0 + 7$

$= 17 = \underline{8}$

Ans $\underline{26.4851}$ ds $\rightarrow$ $\underline{8}$

(12) From his monthly income X spends 24% on household expense, 16% on entertainment, 28% on education out of remaining he donates 72% money & will be left with 2403.52 find his monthly income.

Sol^n

$x \rightarrow$ person.

24% } spending
16% }
28%
68% $\rightarrow$ spending.

In that the remaining is 32%

& also donate the 72%

means remaining 28%

$\therefore$ 28% = 2403.52

= 7

$\therefore x$ of $\underset{+}{32}\%$ of $28\% = 2403.82$

$x \times \cancel{5} \times 10 = 7$

$x \times \cancel{5} \times 1 = 7$

$x = \frac{7}{5}$

$x = \frac{7 \times 2}{5 \times 2} = \frac{14}{10} = \frac{5}{1} = \underline{5}$

(d.s.)

<u>Ans</u> (b) $\underset{+\,+\,+\,+}{.26825} \to \underset{+}{14} = \underline{5}$ <u>d.s</u>

(13) Jeetu buys 14 sarees at an avg cost of Rs. 835. If he buys 11 sarees more at avg cost of ₹ 645. What will be the avg cost of all sarees he bought?

<u>Sol^n</u>

$14 \to 835$

$11 \to 645$

<u>for avg</u>

$= \frac{14 \times 835 + 11 \times 645}{(14 + 11)}$

$= \frac{5 \times \overset{1,6}{7} + 2 \times 6}{5 + 2}$ d.s.

$= \frac{35 + 12}{7} = \frac{8 + 3}{7}$

$= \frac{11}{7} = \frac{2}{7} = \frac{2 \times 4}{7 \times 4} = \frac{8}{28} = \frac{8}{10}$

$= \frac{8}{10} = \frac{8}{1}$.

$= 8$ (digital sum)

Ans = (a) 751.40  8 (ds)

(14) What is the value of

$-15 + 90 \div [89 - \{9 \times 8 + (33 - 3 \times 7)\}]$

Sol^n

$= -15 + 0 \div$ Any thing is 0

$= -15 + 0$

$= -(6) + 0$.

$= -6 + 9$ (because digital sum by 9)

$= 3$ (ds)

Ans = (c) 3.